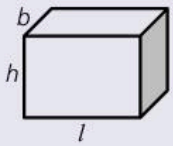
5.4/5.5 Mensuration

You need to:

- Carry out calculations involving the surface area and volume of a cuboid, prism and cylinder.
- Carry out calculations involving the surface area and volume of a sphere, pyramid and cone.
- Carry out calculations involving the areas and volumes of compound shapes.

Recap

Cuboid

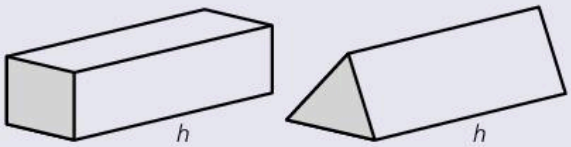


Volume of cuboid = $b \times l \times h$

Surface area of cuboid = $2hl + 2hb + 2bl = 2(hl + hb + bl)$

Prism

Volume of a prism = cross-sectional area × height



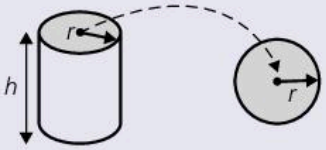
Watch out!

There is no formula as such for the surface area of a prism. It depends on the shape and dimensions of the cross-section.

Recap

Cylinder

A cylinder is a prism with a circular cross-section.



Cross-sectional area = πr^2
Volume of cylinder = $\pi r^2 h$

The **curved surface area** of a cylinder = $2\pi rh$ (or πdh where d is the diameter)

The **total surface area** of a cylinder = $2\pi rh + 2\pi r^2$

Watch out!

Make sure you read the question carefully to ensure that you calculate the correct surface area.

Key skills

You need to be able to calculate the volume of a cylinder in context.

Exam tip

The formula for volume requires the *radius* of the cylinder.

Worked example

A cylindrical can of soup has a diameter of 7.5 cm and a height of 11 cm.
Calculate the volume of the can. **[2 marks]**

Radius = 3.75 cm
Volume = $\pi \times 3.75^2 \times 11 = 485.965\ldots$
$= 486\ \text{cm}^3$ (to 3 s.f.)

Worked example

A cylindrical can of soup has a diameter of 7.5 cm and a height of 11 cm. Calculate:

(a) the curved surface area of the can

(b) the total surface area of the can. **[4 marks]**

(a) Curved surface area $= \pi \times 7.5 \times 11$
$= 259.181\ldots$
$= 259\ \text{cm}^2$ (to 3 s.f.)

(b) Total surface area = curved surface area $+ 2\pi r^2$
$= 259.181\ldots + 2 \times \pi \times 3.75^2$
$= 347.538\ldots$
$= 348\ \text{cm}^2$ (to 3 s.f.)

Key skills

You need to be able to calculate the surface area of a cylinder in context.

Exam tip

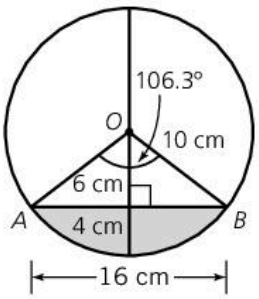
Use the given diameter to calculate the curved surface area.

Exam tip

Use the exact value of the curved surface area so as not to introduce rounding errors.

Worked example

The cross-section of a water pipe is a circle of radius 10 cm. Water flows along the pipe at a depth of 4 cm, as shown in the diagram.



$OB = 10$ cm, $AB = 16$ cm and angle $AOB = 106.3°$

Given that the water is moving at 1.2 m per minute, calculate the volume of water which flows along the pipe in one hour.

Give your answer to 3 significant figures. **[6 marks]**

Area of cross-section of water = area of minor segment bounded by chord AB

Area of minor segment = area of minor sector – area of triangle AOB

Area of minor sector $= \frac{106.3}{360} \times \pi \times 10^2 = 92.729\ldots\ \text{cm}^2$

Area of triangle $AOB = \frac{6 \times 16}{2} = 48\ \text{cm}^2$

Hence area of cross-section of water $= 92.729\ldots - 48 = 44.729\ldots\ \text{cm}^3$

Rate of flow = 120 cm per minute

Volume of water in 1 minute $= 44.729\ldots \times 120 = 5371.71\ldots\ \text{cm}^3$

So in 1 hour, volume of water $= 5371.71\ldots \times 60 = 322\,000\ \text{cm}^3$ (to 3 s.f.)

The region shaded grey in this diagram is a segment of the circle. A segment of a circle is the region bounded by a chord and an arc.

Watch out!

6 marks indicate that this is going to be a multi-stage solution, so make sure you set your working out neatly and logically. You might want to work in 'rough' first.

Exam tip

Write down what you intend to calculate and how you are going to get there, then work the calculation through.

Exam tip

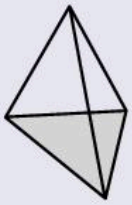
Avoid using a rounded value for the cross-sectional area of the water. This ensures that your final answer will be correct when rounded.

Watch out!

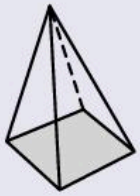
There is no formula as such for the surface area of a pyramid. It depends on, among other things, the shape and dimensions of the base.

Recap

Pyramid



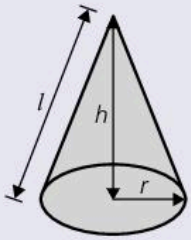
Triangular pyramid or tetrahedron



Square-based pyramid

Volume of pyramid $= \frac{1}{3} \times$ base area $\times$ height

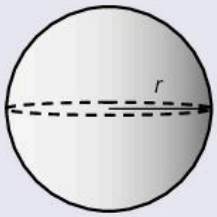
Cone



Volume of cone $= \frac{1}{3}\pi r^2 h$

The **curved surface area** of a cone $= \pi r l$ where l is the 'slant height'.

Sphere



Volume of sphere $= \frac{4}{3}\pi r^3$

Surface area of sphere $= 4\pi r^2$

Key skills

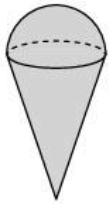
You need to be able to work out the volume of compound solids. The formulae for the surface area and volume of a sphere, cone and pyramid will be given in the question when required. You do not need to learn them.

Exam tip

Work out the two volumes separately. Don't forget to halve the formula for the volume of a sphere.

Worked example

An ice cream is modelled as a cone with a hemisphere on top. The height of the cone is 10 cm and the radius of the cone is 3 cm. Find the total volume of the ice cream, giving your answer as an exact multiple of π. **[4 marks]**



Volume of cone $= \frac{1}{3} \times \pi \times 3^2 \times 10 = 30\pi \text{ cm}^3$

Volume of hemisphere $= \frac{\frac{4}{3} \times \pi \times 3^3}{2} = 18\pi \text{ cm}^3$

Total volume $= 30\pi + 18\pi = 48\pi \text{ cm}^3$

Questions

1 A cylindrical glass of radius 4 cm and height 9 cm is filled with water. The water is then poured into an upturned cone of base radius 5 cm and height 15 cm until the cone is full. How much water (to 2 s.f.) will be left in the glass?

2 A company makes spherical and cubical ice holders. What is the radius (to 3 s.f.) of the spherical container if it has the same volume as the cubical container with a side length 13 mm?

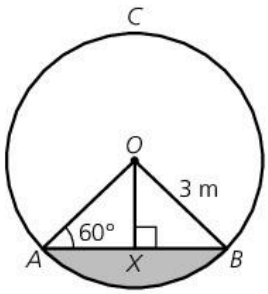
3 An ice cream scoop is designed to make spheres. Ice cream is taken from a container measuring 20 cm by 15 cm by 13 cm. If the scoop always picks up perfect spheres of radius 2.4 cm, how many scoops (to the nearest whole number) can be filled from the container (assuming no waste).

4 A 3-litre pot of paint is used to cover the surface of a large sphere. The instructions say that one litre of paint will cover 5 m^2. What is the maximum radius of the sphere if it is to be completely covered?

5 A cone of height 45 cm has to hold at least 3 litres of water. What is the least possible value (to 3 s.f.) of the base radius?

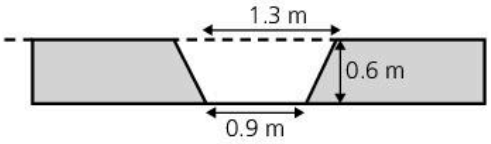
6 A railway tunnel is constructed in the shape of a hollow cylinder. It is 1 km long and has a radius of 3 m. A gravel bed is laid in order to support the track. A cross-section of the tunnel is shown in the diagram, with the shaded area representing the gravel. AB represents the horizontal surface of the gravel.

X is the midpoint of AB and $\angle OAB = 60°$.



a Show that $OX = 2.60$ m (to 3 s.f.).
b Find the area (to 3 s.f.) of the triangle OAB.
c By considering the area of the sector OAB and your answer to part **b**, find (to 3 s.f.) the shaded area.
d Hence find (to 3 s.f.) the volume of gravel required for the tunnel.
e Find (to 3 s.f.) the length of the major arc ACB.
f The wall of the tunnel above the gravel level is to be painted. Find (to 3 s.f.) the surface area to be painted.

7 A ditch is cut in the ground in such a way that its cross-section is a trapezium as shown in the diagram.



The ditch is 400 m long and it is filled with water.

a What is the area of the cross-section of the ditch?
b What is the volume of water which the ditch contains?
c The water flows at a rate of 1.2 m s^{-1}. What volume of water passes one point in a minute?

A pipe with a square cross-section is then placed in the ditch and the rest of the ditch is filled in with soil.

d If the pipe has the largest possible cross-sectional area, how much soil is put back into the ditch?

8 The Earth is roughly a sphere with a radius of 6371 km. If about 30% of the Earth's surface is land, roughly what area is this in km^2? Give your answer correct to 3 significant figures.

To **Raise your grade** now try questions 1, 3 and 4, page 142